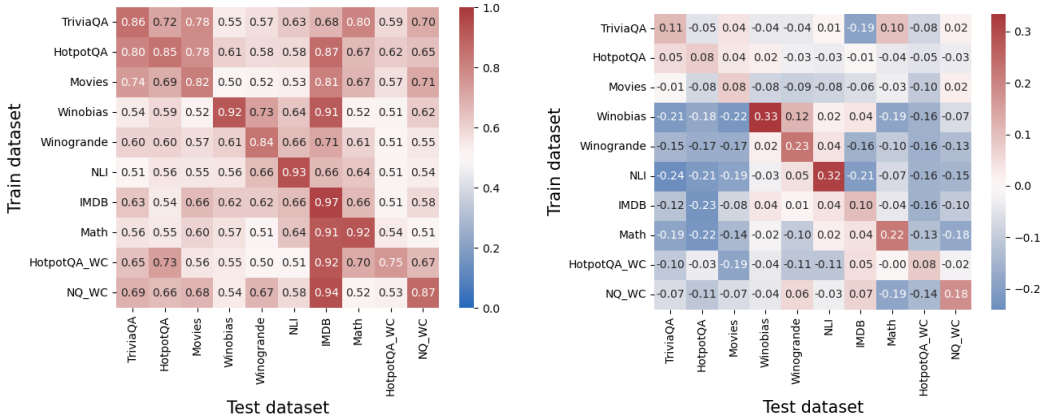


(a) Raw AUC values. Values above 0.5 indicate some generalization.

(b) Performance (AUC) difference of the probe and the logit-based method. Values above 0 indicate generalization beyond the logit-based method.

Figure 3: Generalization between datasets, Mistral-7b-instruct. After subtracting the logit-based method's performance, we observe that most datasets show limited or no meaningful generalization.

which only observes the output logits. Figure 3b presents the same heatmap after subtracting results from our strongest logit-based baseline (Logit-min-exact). This adjusted heatmap reveals the probe's generalization rarely exceeds what can be achieved by examining logits alone. This suggests that the observed generalization is not due to a universal internal encoding of truthfulness. Instead, it likely arises from information already available through external features, such as logits. Past evidence for generalization may therefore have been overstated.

Nonetheless, we do observe some successful generalization in tasks requiring similar skills, such as parametric factual retrieval (TriviaQA, HotpotQA, Movies) and common-sense reasoning (Winobias, Wingrande, NLI). This suggests that, although the overall pattern of truthfulness signals across tokens appeared consistent across tasks (as observed in Section 3.3), LLMs have many "skill-specific" truthfulness mechanisms rather than universal ones. However, some patterns remain unexplained, such as the asymmetric generalization from TriviaQA to Math tasks. Overall, our findings indicate that models have a multifaceted representation of truthfulness. The internal mechanisms responsible for solving distinct problem are implemented as different mechanisms (e.g., circuits) within models (Elhage et al., 2021; Olah et al., 2023). Similarly, LLMs do not encode truthfulness through a single unified mechanism but rather through multiple mechanisms, each corresponding to different notions of truth. Further investigation is required to disentangle these mechanisms.

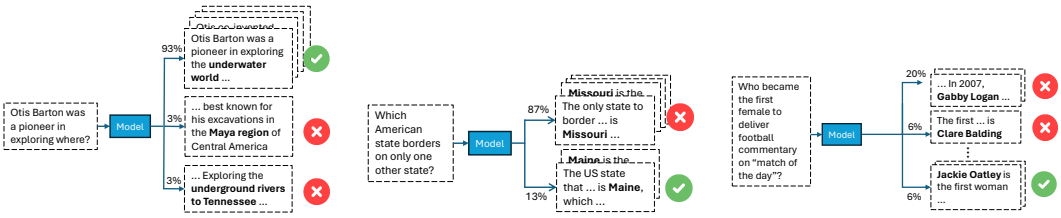
5 Investigating Error Types

Having established the limitations of error detection, we now shift to error analysis. Previously, we explored types of LLM limitations across *different* tasks, noting both commonalities and distinctions in their error representations. In this section, we focus on the types of errors LLMs make in a *specific* task—TriviaQA—which represents factual errors, a commonly studied issue in LLMs (Kadavath et al., 2022; Snyder et al., 2023; Li et al., 2024; Chen et al., 2024; Simhi et al., 2024).

5.1 Taxonomy of Errors

Intuitively, not all mistakes are identical. In one case, an LLM may consistently generate an incorrect answer, considering it correct, while in another case, it could issue a best guess. To analyze errors from the LLM's perspective, we sample $K = 30$ responses at a temperature setting of $T = 1$[4] for each example in the dataset and then analyze the resulting distribution of answers.

[4]We chose $K = 30$ as the overall correctness seemed to plateau around this point; see Appendix D. We found that lower temperatures generally produced less truthful answers across repeated trials.



(a) The LLM mostly answers correctly, but sometimes hallucinates.

(b) The LLM mostly answers incorrectly, but seems to have some knowledge on the correct answer.

(c) The LLM generates many different answers, one of them is the correct one which is generated a small fraction of the resamples.

Figure 4: Different error types in free-form generation, exposed when resampled many times.

Figure 4 illustrates three representative error types. In one (Figure 4a), the model usually gives the correct answer but occasionally make an error, implying correct information is present but sampling may lead to mistakes. In another (Figure 4b, the model often responds incorrectly, though it is capable of providing the right answer, indicating some retained knowledge despite consistently making the same error. In a third type (Figure 4c), the model generates a wide array of mostly incorrect answers, reflecting low confidence in any generated answer.

More generally, we categorize the errors by logging three specific features for each example: (a) the number of different answers generated; (b) the frequency of the correct answer; and (c) the frequency of the most common incorrect answer. These features reveal the following error patterns:

- **(A) Refuses to answer:** The model responds that it cannot answer the question in at least half the cases.
- **(B) Consistently correct:** Answers correctly in at least half of the cases. This category is divided into: (B1) always correct; and (B2) mostly correct with occasional errors.
- **(C) Consistently incorrect:** Consistently generates the same incorrect response in at least half of the cases. Similarly to type B, we subdivide this type into (C1) correct answer is never produced; and (C2) correct answer appears at least once.
- **(D) Two competing:** Generates both correct and incorrect responses at similar rates– difference in rates is 5 or less, and each response is generated at least 5 times.
- **(E) Many answers:** Generates over 10 distinct answers. Like types C and D, Subtypes include (E1) correct answer is never generated; and (E2) correct answer is generated at least once.

This taxonomy covers 96% of the errors in TriviaQA for Mistral-7b-instruct. For more qualitative examples of each type of error, see Appendix D.3. Although some overlap exists between types, our goal is to identify general patterns and explore their connection to the models's internal representations. For a discussion on the design choices of this taxonomy, refer to Appendix D.1. This taxonomy classifies LLM errors based on an extrinsic, behavior-based analysis. Similarly, previous work analyzed repeated samples to assess an LLM's knowledge of the correct answer (Simhi et al., 2024; Gekhman et al., 2024). Our approach is distinct because it also examines the nature of errors that the LLM makes. Furthermore, as we discuss next, we analyze the connection between these behavioral patterns and the model's internal encoding.

5.2 Predicting Error Types

Our taxonomy offers an external, behavioral analysis of LLMs, which we complement by an intrinsic evaluation. We explore whether LLMs encode information on potential error types within their intermediate activations, offering a deeper insight into the underlying mechanisms. To investigate this, we train a probe in a one-to-many setting, where a single probe identifies a specific error type from all others. We use representations extracted from the answers produced via greedy decoding.

Table 2 presents the results. Our findings show that error types can be predicted from the intermediate representations of the greedy decoding generations, suggesting that they may capture not only

Table 2: AUC scores for error type classification (TriviaQA). Error types are predictable from the inner model representations, indicating the encoding of fine-grained information on errors.

Error type	Mistral-7b	Mistral-Instr-7b	Llama3-8b	Llama3-Instr-8b
(A) Refuses to answer	$0.86_{\pm 0.002}$	$0.85_{\pm 0.011}$	$0.87_{\pm 0.002}$	$0.88_{\pm 0.014}$
(B) Consistently correct	$0.88_{\pm 0.001}$	$0.82_{\pm 0.008}$	$0.86_{\pm 0.001}$	$0.81_{\pm 0.002}$
(C) Consistently incorrect	$0.59_{\pm 0.002}$	$0.67_{\pm 0.002}$	$0.59_{\pm 0.002}$	$0.64_{\pm 0.003}$
(D) Two competing	$0.63_{\pm 0.002}$	$0.68_{\pm 0.006}$	$0.61_{\pm 0.001}$	$0.65_{\pm 0.004}$
(E) Many answers	$0.90_{\pm 0.001}$	$0.84_{\pm 0.003}$	$0.89_{\pm 0.001}$	$0.89_{\pm 0.001}$

output correctness but also fine-grained information about potential errors. While detection performance varies between types, the predictability of each type is valuable on its own, as it opens the possibility of tailoring targeted interventions for specific error types. Additionally, although performance on error types C and D is lower, it remains well above random, providing meaningful insights. These results suggest that internal representations encode more than just binary correctness, revealing a nuanced taxonomy of error types and offering deeper insights into how these models process and encode knowledge.

6 Detecting the Correct Answer

After identifying that models encode diverse truthfulness-related information, we examine how this *internal* truthfulness aligns with their *external* behavior during response generation. To this end, we use our probe,[5] trained on error detection, to select an answer from a pool of 30 generated responses to the same question. We then measure the model's accuracy based on the selected answers. A case where this accuracy does not significantly differ from traditional decoding methods (such as greedy decoding), suggests that the LLM's internal representation of truthfulness is consistent with its external behavior. In simpler terms, that the model is generating answers that it also internally considers as correct. Conversely, a case where using the probe alters performance either way, would suggest a misalignment between the LLM's internal representations and its actual behavior.

Experimental Setup The experiments were conducted on TriviaQA, Winobias, and Math. We resample each model answer in the same strategy described in Section 5.1. The final chosen answer is the one with the highest correctness probability, as assessed by the probe. We compare to three baselines: (1) greedy decoding, (2) random selection from the $K = 30$ answer candidates; and (3) majority vote wherein the most frequently generated answer is chosen.

Results The results for Mistral-7b-instruct are summarized in Figure 5, with additional results for other LLMs and datasets as well as qualitative examples provided in Appendix E. We only present results on error types that appear 30 times or more in our test dataset. Overall, using the probe to select answers enhances the LLMs accuracy across all examined tasks. However, the extent of improvement varies by error type. For instance, in the TriviaQA dataset, there is minimal gain in the "mostly correct" category (B2). In contrast, substantial gains—ranging from 30 to 40 points in some cases—are observed in the "mostly incorrect" (C2), "two competing answers" (D), and "many answers" (E1) categories. Interestingly, and perhaps surprisingly, the probe is most effective in cases where the LLM lacks any (external) preference for the correct answer during generation. The fact that the probe can effectively identify the correct answer in these scenarios, points at a significant disconnect between the LLM's internal encoding and its external behavior. These results suggest that even when the model encodes information of which answer is correct, it can still generate an incorrect answer in practice.

While using the probe to select the answer proves effective, it is not proposed here as an error mitigation strategy but rather as a diagnostic tool. However, these findings indicate that further research in this area could leverage the existing knowledge within LLMs to significantly reduce errors. We recommend exploring this direction in future investigations.

[5]We choose the best-performing probe for each task, which is trained on the last exact answer token.